Sahaj Bhadja

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Summary

Software Engineer with 2+ years building maintainable **middleware**, **REST APIs**, and scalable cloud applications using OOP, Java, Python, TypeScript, Azure, SQL, microservices, testing, and Agile practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected a multi-tenant platform using Azure SQL and Blob Storage with row-level isolation and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built scalable middleware across FastAPI, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and CI/CD releases sustaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Implemented production Python services on AWS supporting 150+ instructors and 10K+ daily requests, using CI/CD-driven workflows to reduce feedback turnaround 60% while maintaining stable API performance under peak load.
- Designed concurrent batch processing and secure multi-format ingestion through REST APIs and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery, usage tracking, and reliable production support.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Designed object-oriented API modules using Singleton and Factory patterns with AWS Glue and S3 transformations, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20%.
- Automated nightly builds, installations, integration tests, unit tests, and reports through Jenkins, validating reliable cross-platform synchronization while collaborating directly with Microsoft and GitHub stakeholders during Agile delivery.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight modular microservices for authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ build validation while integrating operational monitoring and analytics across controlled staging and production environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Developed Flask APIs, payment webhooks, pricing, and booking workflows while synchronizing Firebase events and processing hourly Databricks micro-batches across 50K+ operational records with consistently reliable sub-200-millisecond responses.
- Designed end-to-end workflows, mapped 12+ integration points, and presented technical solutions to client stakeholders, reducing design errors 25% while supporting effective collaborative cross-functional implementation planning and alignment.

Skills

Languages & Frameworks: Java, Python, C++, TypeScript, JavaScript, SQL, React, FastAPI, Flask, Node.js
Middleware & Data: OOP, REST APIs, Microservices, WebSockets, Authentication, Authorization, Database Design, PostgreSQL, SQL Server, Azure SQL, MongoDB
Cloud & Delivery: Azure, AWS, S3, Blob Storage, Docker, Kubernetes, Jenkins, CI/CD, Git, GitHub, Postman, Linux
Engineering: Agile/Scrum, Code Reviews, Pull Requests, Unit Testing, Integration Testing, Debugging, Monitoring, Requirements Analysis, Documentation, Stakeholder Communication

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, WebSockets*

- Built a scalable application supporting 50K+ users and 1M+ transactions with secure third-party integrations, real-time synchronization, token-based access controls, advanced filtering, and reliable Kubernetes deployment workflows globally.

HMIS - Hospital Management Information System Live | *TypeScript, MERN, REST APIs, Sentry*

- Implemented consent-based access, role-oriented workflows, REST integrations, automated alerts, and Sentry monitoring through scalable TypeScript services supporting 100+ daily operations while reducing hospital administrative workload by 30%.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built a Java and TypeScript database workspace supporting 10+ systems, accelerating query authoring, schema exploration, and data management through reusable backend services and a unified cross-platform interface.